Beyond Semantic Confidence: Relation-Consistent Geometric Re-ranking for 3D Scene Graphs

Anonymous submission

Abstract

3D scene graph predictors can assign high scores to relation candidates whose predicates are inconsistent with the geometry of the reconstructed ordered object pair. We formulate predicate–geometry compatibility assessment for fixed relation predictions and introduce RelCompat3D, which preserves ordered-pair identity and separates predicate/family semantics, predicate-independent geometric measurements of the ordered pair, and the relation score produced by the predictor. Family-specific compatibility heads are trained using linked positive–counterfactual pairs without predictor identity or the predictor score. Transformation averaging enforces proximity symmetry and invariance under joint endpoint swap and inverse-predicate transformation. The method re-ranks proximity and vertical relations while keeping support/contact relations in source order. We evaluate Recall@K with exact predicate matching jointly with Violation@K, computed by a rule-based geometry verifier, across VL-SAT, Open3DSG, and SceneGraphFusion (SGFN) on a shared 3DSSG target. Across the reported budgets, RelCompat3D yields lower or tied Violation point estimates while preserving or improving Recall point estimates, with the largest gains on Open3DSG. Comparisons with matched rank-fusion and nonlinear baselines reveal predictor- and family-dependent trade-offs.

Introduction

Figure 1 shows a pair-level Top-50 demotion; Table 1 reports aggregate results.

3D scene graphs turn reconstructed scenes into structured object-relation representations that can support spatial reasoning, embodied AI, scene alignment, visual grounding, and downstream decision making (Wald et al. 2020; Sarkar et al. 2023; Xie, Pagani, and Stricker 2024; Liu et al. 2026). For these uses, relation prediction must be supported by the reconstructed scene, not only plausible at the category or language level. A graph edge such as `standing on`, `close by`, or `higher than` is useful only if it describes the corresponding ordered subject–object pair in 3D.

We study a failure mode in which relation predictors rank plausible labels that are inconsistent with the measured geometry of the ordered object pair. A relation can sound plausible for two object categories while being inconsistent with measured distance, projected overlap, or vertical arrangement; relations requiring contact evidence are assessed separately by the verifier. This mismatch becomes consequential

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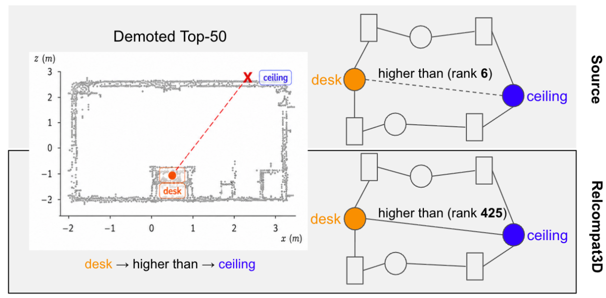

Figure 1: Open3DSG demotion example. The geometrically inconsistent *desk higher than ceiling* relation moves from rank 6 to 425 after RelCompat3D.

as 3D scene graphs become interfaces for open-vocabulary language-based applications and embodied systems (Koch et al. 2024b; Hou et al. 2025; Zhang et al. 2025; Wang et al. 2025; Saxena and Chiun 2025; Yu et al. 2025; Madhavaram et al. 2026).

The failure is not simply that existing 3DSSG methods ignore geometry. Prior work already uses edge features, point-cloud structure, spatial knowledge, multimodal semantics, and geometric fusion (Zhang et al. 2021; Feng et al. 2023; Wang et al. 2023; Koch et al. 2024a; Chen et al. 2024). The gap is more specific: a predictor score—even when its predictor already uses geometry—is not necessarily an explicit estimate of whether the same subject-object pair satisfies the predicate in 3D. This makes the problem an AI reliability issue, not only a 3DSSG implementation detail: language-based and embodied systems can inherit a high-scoring symbolic relation that contradicts the reconstructed scene evidence.

This failure suggests separating predicate semantics T, predicate-independent geometric measurements of the ordered pair G, and the predictor score Z. RelCompat3D preserves each candidate's subject–object identity and learns predicate–geometry compatibility from T and G without using Z or predictor identity. Linked positive–counterfactual pairs optimize the ordering of geometrically compatible and incompatible candidates; averaging over known endpoint/predicate transformations enforces proximity symmetry and invariance under joint endpoint swap and inverse-

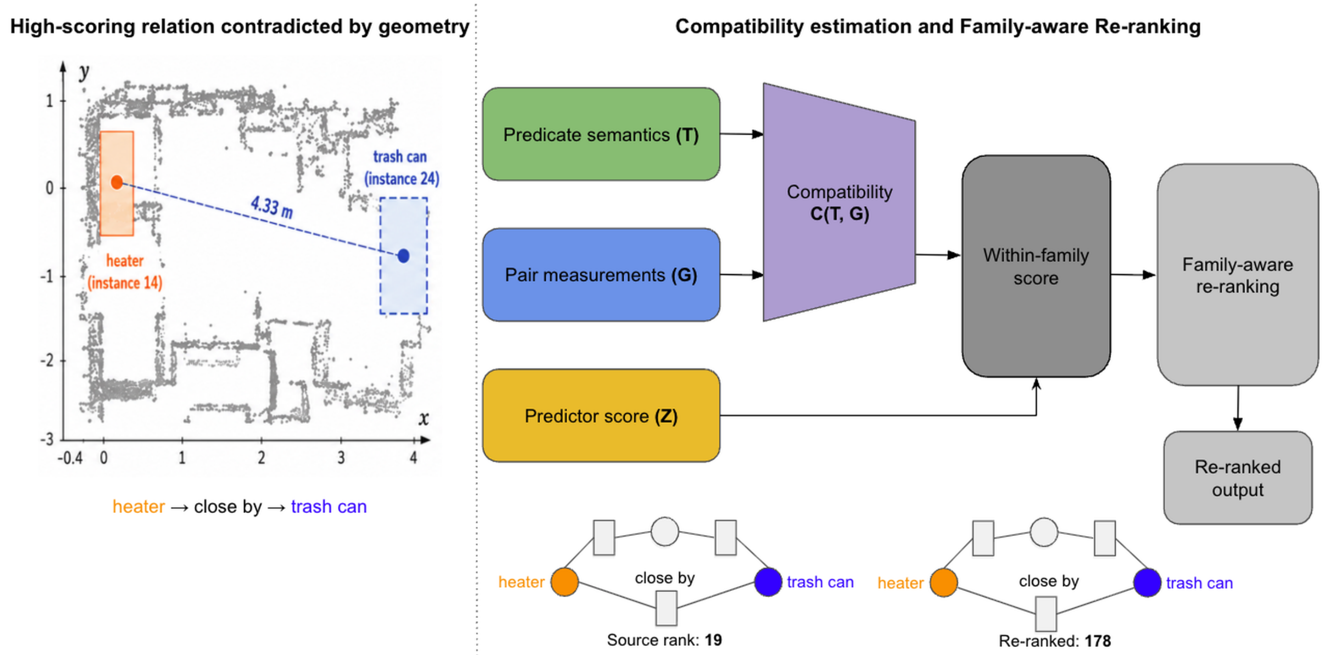

Figure 2: RelCompat3D overview. Predicate semantics T and predicate-independent geometric measurements G determine compatibility without the predictor score Z. After transformation averaging, Z enters the within-family score used for family-aware re-ranking; the shown incompatible proximity relation moves from rank 19 to 178.

predicate transformation. These choices address distinct failure modes: excluding Z precludes direct copying of the predictor score, preserving ordered-pair identity prevents scene-level measurements from substituting for instance evidence, and transformation averaging assigns equal compatibility to equivalent endpoint/predicate representations.

The predictor score enters only during re-ranking. Within proximity and vertical relations, RelCompat3D combines it with compatibility and changes the order within each relation family. It preserves the sequence of relation-family labels in the source ranking and the source order of support/contact candidates. Matched fusion baselines and ablations test the effects of excluding the source predictor score, preserving ordered-pair identity, and enforcing transformation consistency. Figure 2 summarizes the framework.

We evaluate support/contact, proximity, and vertical-order relations because their consistency can be assessed from the geometry of reconstructed ordered object pairs. Open3DSG, VL-SAT, and SceneGraphFusion (SGFN) provide complementary open-vocabulary and closed-set predictors (Wang et al. 2023; Koch et al. 2024b; Wu et al. 2021). We jointly report Recall@K with exact predicate matching and Violation@K computed by a rule-based geometry verifier across the three predictors on a shared 3DSSG target.

Our contributions are threefold:

1. We define and measure a mismatch between relation scores and instance-level geometric consistency that exact-match Recall alone does not capture, using identity-preserving evaluation and joint Recall@K–Violation@K reporting.
2. We learn predicate–geometry compatibility from the corresponding ordered pair without using the predictor score, and enforce exact endpoint/predicate transformations through transformation averaging.
3. We introduce a constrained re-ranking rule that preserves the source relation-family sequence and support/contact candidates in source order, and evaluate one fitted compatibility layer with family-specific heads across three predictors to characterize predictor- and family-dependent gains and failures.

Across $K = 10$–50, no predictor's Recall point estimate decreases, and at $K = 50$ paired intervals from a cluster bootstrap over scans support lower Violation without a detectable Recall loss. RelCompat3D operates on fixed relation predictions rather than replacing the relation generator.

Related Work

3D scene graph relation prediction. 3D scene graphs organize indoor scenes as object-relation structures grounded in 3D space (Armeni et al. 2019). The 3RScan/3DSSG setting introduced a benchmark and evaluation setting for 3D relation prediction (Wald et al. 2020), while incremental and online systems study how scene graphs can be constructed from RGB-D streams or spatial perception stacks (Wu et al. 2021; Hughes, Chang, and Carlone 2022). Recent predictors improve closed-set predicates using edge reasoning, point-cloud features, multimodal cues, spatial knowledge, self-supervised pretraining, object-centric features, and edge-centric relational reasoning (Zhang et al. 2021; Feng et al. 2023; Wang et al. 2023; Koch et al. 2024a; Heo et al.

2025; Ma et al. 2026). These methods already use geometry; RelCompat3D asks the complementary question of whether a fixed high-scoring predicate is geometrically compatible with the corresponding reconstructed ordered pair.

Open-vocabulary 3D graphs. Open-vocabulary 3D perception provides queryable object and region features (Peng et al. 2023; Takmaz et al. 2023); graph systems then connect such features to compact scene representations for querying, planning, navigation, online mapping, and language interaction (Gu et al. 2024; Werby et al. 2024; Maggio et al. 2024). Recent 3D scene graph work extends this line toward open-vocabulary objects, open-set relations, VLM features, online graph generation, and functional relations (Koch et al. 2024b; Chen et al. 2024; Hou et al. 2025; Zhang et al. 2025; Wang et al. 2025). This line raises the reliability requirement: when relation edges become scene representations consumed by language-based systems, it is important to assess whether plausible text labels are consistent with the reconstructed geometry of the corresponding ordered pair.

Geometry-aware relation evidence. 3D relation prediction and graph reuse already rely on edge features, metric distance, topology, and semantic-geometric fusion (Zhang et al. 2021; Feng et al. 2023; Xie, Pagani, and Stricker 2024). Adjacent work on object placement and affordance grounding also uses support/contact patterns and affordance geometry (Huang et al. 2025; Shao et al. 2025). Scene graph alignment and registration methods further demonstrate that graph structure and geometric consistency matter when scene graphs are reused for matching, alignment, and downstream geometric tasks (Sarkar et al. 2023; Xie, Pagani, and Stricker 2024).

RelWitness integrates visual-geometric witnesses into open-vocabulary 3D scene graph learning and decoding (Nguyen et al. 2026). Statistical confidence rescoring learns source-specific refinement from semantic masks, neighboring relations, and statistical priors (Yeo, Li, and Lee 2025). PUF introduces training-free uncertainty-aware fusion for online scene-graph association and evidence accumulation (Yang et al. 2026). RelCompat3D instead evaluates fixed relation candidates using predicate–geometry compatibility whose inputs exclude predictor identity and the predictor score.

Several recent systems incorporate geometry or transformation structure directly into relation generation. GEODE conditions predicate generation on updated geometry tokens inside a discrete-diffusion generator (Feng et al. 2026), while RelGraphOV prunes geometrically implausible connections when learning an open-vocabulary scene representation (Chen et al. 2026). Transformation-Aware Decoupling separates predicate reasoning according to how labels transform under yaw-viewpoint changes inside the generator (Sun et al. 2026). In contrast, RelCompat3D leaves each predictor fixed. After prediction, it averages compatibility over applicable joint transformations of the predicate and the geometry of the ordered pair, while excluding the predictor score from compatibility inputs.

Neau et al. mine spatial, functional, and relation-algebra constraints and use declarative reasoning to refine fixed image-SGG outputs, reporting a Constraint Violation Rate alongside task metrics (Neau et al. 2026). RelCompat3D differs in estimating continuous compatibility from reconstructed 3D geometry, preserving ordered-pair identity, and enforcing applicable endpoint/predicate transformations before joint Recall@K–Violation@K evaluation.

Reliability evaluation. Standard Recall@K asks whether the correct predicate appears near the top, not whether the predicted relation is geometrically consistent with the reconstructed scene. Confidence calibration asks whether model scores reflect empirical correctness likelihood (Guo et al. 2017). RelCompat3D instead associates fixed relation predictions with 3D evidence from the corresponding ordered subject–object pair and reports Recall@K with exact predicate matching together with Violation@K computed by a rule-based geometry verifier. Its compatibility output is a bounded ranking score, not a probability of physical validity.

Method

RelCompat3D assesses fixed relation predictions rather than replacing a 3D Scene Graph generator. It preserves candidate identity, estimates predicate–geometry compatibility without the predictor score, and combines the two only during reranking (Figure 2).

Problem Setup

Let a relation predictor produce relation candidates

$$r_i = (\mathrm{scan}_i, \mathrm{context}_i, s_i, o_i, p_i, a_i, Z_i), \tag{1}$$

where s_i and o_i are ordered subject and object instance identifiers, p_i is the predicate, a_i its mapped relation family, and Z_i the relation score produced by the predictor. A context is a scene subgraph over which the predictor ranks candidates; top-K selection is performed independently within each context. The tuple $(\mathrm{scan}_i, \mathrm{context}_i, s_i, o_i)$ uniquely identifies the ordered object pair and is retained when associating each prediction with geometry and ground truth. We represent predicate/family semantics as $T_i = (p_i, a_i)$ and associate predicate-independent geometric measurements of the ordered pair

$$G_i = G(\mathrm{scan}_i, \mathrm{context}_i, s_i, o_i). \tag{2}$$

The measured scope is

$$\mathcal{A} = \{\mathrm{support/contact},\ \mathrm{proximity},\ \ \mathrm{vertical\ order}\}, \tag{3}$$

whose consistency can be assessed from the geometry of the reconstructed ordered object pair. Candidates outside these families are excluded from the measured violation claim. The re-ranking scope is narrower: $\mathcal{A}_{\mathrm{tr}} = \{\mathrm{proximity}, \mathrm{vertical\ order}\}$, for which the defined endpoint/predicate transformations are exact. Support/contact is evaluated but kept in source order because no single endpoint transformation preserves the semantics of every predicate in this family.

Our objective is to estimate a bounded compatibility score

$$C_i = \sigma(h_{a_i}(\Phi(T_i, G_i))) \in [0, 1], \tag{4}$$

then average over the valid family-specific endpoint/predicate transformations to obtain the transformation-consistent score C_i^{tr} and form a within-family ranking score $u_i = F(Z_i, C_i^{\mathrm{tr}})$. Training labels pair ground-truth relations with counterfactual negatives formed by reversing a predicate or replacing an endpoint when the geometry strongly contradicts the label (full rules and thresholds are in the supplement). The resulting score is not a probability of physical validity. Features that combine the predicate with predicate-independent pair measurements belong to the $T_i \times G_i$ term. Neither the predictor score nor predictor identity enters C_i. The same fitted parameters are therefore used for all three predictors on the shared target.

Let $L_K(c)$ be the top-K candidates from the evaluated relation families for context c and Y_c its exact-match ground-truth set. We jointly report

$$\begin{aligned} \text{Recall@K} &= \frac{\sum_c |L_K(c) \cap Y_c|}{\sum_c |Y_c|}, \\ \text{Violation@K} &= \frac{1}{|D_K|} \sum_{(c,i)\in D_K} \mathbf{1}[v_i = \text{violated}]. \end{aligned} \tag{5}$$

where D_K contains selected candidates for which the verifier returns $v_i \in \{\text{satisfied}, \text{uncertain}, \text{violated}\}$. Family mapping never relaxes exact predicate matching. Because uncertain candidates enter the denominator, the experiments also report violation over the decidable subset, pessimistic violation, uncertainty, and coverage.

Relation-Consistent Compatibility

For each candidate relation r_i, the associated geometry vector G_i contains OBB-derived 3D/XY distances, relative height, projected overlap, and vertical-gap features. Point-level contact evidence is used only by the support/contact verifier, not by the compatibility model. The feature map $\Phi = (\phi_T, \phi_G, \phi_\times)$ separates predicate/family indicators, predicate-independent geometry, and explicit predicate–geometry interactions. The predictor score, rank, predictor identity, and object-class features are excluded.

We first define a family logit and bounded compatibility score

$$\begin{aligned} h_a(\Phi(T_i, G_i)) &= w_a^\top \Phi(T_i, G_i), \\ C_i &= \sigma(h_a(\Phi(T_i, G_i)))\,. \end{aligned} \tag{6}$$

For proximity, τ swaps the ordered endpoints and leaves `close by` unchanged. For vertical order, τ swaps the endpoints and maps `higher than`$\leftrightarrow$`lower than`. After transformation averaging, the final score satisfies

$$\begin{aligned} C^{\mathrm{tr}}_{\mathrm{close}}(s, o) &= C^{\mathrm{tr}}_{\mathrm{close}}(o, s), \\ C^{\mathrm{tr}}_{\mathrm{higher}}(s, o) &= C^{\mathrm{tr}}_{\mathrm{lower}}(o, s). \end{aligned} \tag{7}$$

Support/contact is evaluated only under the identity transformation because no single endpoint swap preserves every predicate in this family. Its compatibility model is used only in the comparison applied to all relation families; the RelCompat3D ranking keeps support/contact candidates in source order.

Training combines transformation-consistent augmentation with a linked positive–counterfactual ranking loss. The augmentation preserves the predicate when proximity endpoints are swapped, whereas vertical order swaps endpoints and applies the inverse predicate. In addition, for every linked positive–counterfactual pair $(i^+, i^-) \in \mathcal{P}$, the logits $\ell = w_a^\top \Phi$ receive a fixed margin penalty:

$$\begin{aligned} \mathcal{L}_a = \mathcal{L}_{\mathrm{BCE}} &+ 0.25\, \mathbb{E}_{\mathcal{P}} \Big[\log\Big(1 + e^{1-(\ell_{i^+} - \ell_{i^-})}\Big) \Big] \\ &+ 10^{-4} \|w_a\|_2^2. \end{aligned} \tag{8}$$

Here $\mathcal{L}_{\mathrm{BCE}}$ is binary cross-entropy over positive and counterfactual examples. Numeric features are standardized and missing values are imputed using statistics estimated from the training split. Each family uses a small linear model fitted exclusively on the training split; optimization details and training-example counts are provided in the supplement.

Finally, let H_{a_i} denote the finite transformation group for family a_i, and define the transformed inputs $\mathcal{O}_i = \{g(T_i, G_i) : g \in H_{a_i}\}$. We average the learned score over this set to obtain the transformation-consistent compatibility:

$$C_i^{\mathrm{tr}} = \frac{1}{|\mathcal{O}_i|} \sum_{(T', G') \in \mathcal{O}_i} \sigma(h_{a_i}(\Phi(T', G')))\,. \tag{9}$$

This transformation averaging makes proximity symmetry and invariance under joint endpoint swap and inverse-predicate transformation exact at inference rather than merely encouraged by a penalty. The pairwise objective still distinguishes each linked positive from its counterfactual; transformation averaging does not use the predictor score.

Proposition 1 (Exact transformation consistency) *Let the finite transformation group H_a act jointly on predicate and geometry of the ordered pair, and define $C_a^{\mathrm{tr}}(x) = |H_a|^{-1} \sum_{g \in H_a} C_a(gx)$. Then $C_a^{\mathrm{tr}}(hx) = C_a^{\mathrm{tr}}(x)$ for every $h \in H_a$. Consequently, Eq. 9 satisfies proximity symmetry and invariance under joint endpoint swap and inverse-predicate transformation for vertical order.*

The supplement gives a proof. Support/contact uses only the identity transformation and is unaffected by this averaging.

Family-Aware Re-ranking

For each candidate, define the within-family ranking score

$$u_i = \begin{cases} Z_i C_i^{\mathrm{tr}}, & a_i \in \mathcal{A}_{\mathrm{tr}}, \\ Z_i, & a_i = \text{support/contact}. \end{cases} \tag{10}$$

For $Z_i, C_i^{\mathrm{tr}} > 0$, $\log u_i = \log Z_i + \log C_i^{\mathrm{tr}}$ in the families with defined endpoint/predicate transformations; compatibility therefore contributes an additive log-score term without an additional fusion parameter. This rule introduces no fitted fusion parameter.

Let $\pi^Z = (e_1, \ldots, e_n)$ denote the source ranking, a_j the relation family at position j, and π_a the candidates of family a sorted by u_i. At position j, the rule selects the next unused candidate from π_{a_j}, exactly preserving the source family-label sequence. Because support/contact candidates are sorted by Z_i, their selected subsequence and family-specific Recall and Violation are unchanged at every K.

For each prefix, the rule also selects the highest-u_i proximity/vertical candidates and maximizes their summed utility subject to the preserved family counts and support/contact subsequence; the supplement gives an exchange proof.

The RelCompat3D product and matched fusion comparators use the same family-aware ranking procedure. Additional controls test the role of family conditioning, the predictor score, pair identity, geometry, and the defined transformations; full definitions are provided in the experiments and supplement.

Experiments

Experimental Setup

Data and splits. Open3DSG, VL-SAT, and SGFN provide complementary open-vocabulary and closed-set relation predictors and enable comparison across predictors on a shared 3DSSG target containing relation families whose consistency can be assessed from the geometry of reconstructed ordered object pairs. All evaluations use the same scope: 157 scans, 548 relation contexts, and 3,972 exact-match ground-truth relations from support/contact, proximity, and vertical-order families. Ground-truth membership is used only when computing Recall@K, never to filter or rank candidate relations.

Open3DSG uses the publicly released model and preprocessing pipeline. Every evaluation context is retained, with unavailable candidate lists treated as empty; coverage and denominator sensitivities are in the supplement.

Baselines and training. We compare the predictor score and RelCompat3D with rank-average fusion (RankAvg), reciprocal rank fusion (RRF), and a matched multilayer perceptron (matched MLP) using the same family-aware re-ranking procedure. Product (all families) tests the effect of re-ranking support/contact, and a pooled compatibility model tests whether family-specific fitting is necessary. The matched MLP uses the same training rows, constructed targets, input features, split, and family-aware ranking procedure as the linear compatibility heads. Controls perturb the predicate, pair identity, geometric measurements of the ordered pair, endpoint order, predictor score, or distance measurements; exact transformations are in the supplement. The compatibility inputs remain limited to predicate and family indicators T, predicate-independent geometric measurements of the ordered pair G, and explicit $T \times G$ interactions. Normalization, imputation, and model parameters use only 1,061 training scans; model design and the applicable relation families were selected on the 117-scan development split. The 157-scan evaluation split is never used for fitting. All comparisons across predictors concern the shared 3DSSG/3RScan target.

A feature-removal analysis refits models exclusively on the training split after excluding either the exact scalar used by the verifier or all related distance or vertical measurements; all conditions are reported in the supplement.

Metrics and uncertainty. Recall@K uses exact predicate matching within each relation context. Violation@K is the fraction of selected candidates assessed as violated by the rule-based geometry verifier. Its denominator includes satisfied, uncertain, and violated candidates, whereas only violated candidates enter the numerator. The decidable-subset variant excludes uncertain candidates from the denominator; the pessimistic variant counts them as violations. We report $K \in \{5, 10, 20, 50, 100\}$ and discuss $K = 50$ as an intermediate reported budget. Paired 95% intervals use a cluster bootstrap over scans: all contexts from a sampled scan are resampled jointly, with shared indices across conditions. Within-family results and family-specific metrics within the selected top-K predictions show whether aggregate changes are driven by relation-family composition.

Recall–Violation Results

Table 1 and Figure 3 show that the family-aware RelCompat3D re-ranking raises the $K = 50$ Recall point estimate and lowers the Violation point estimate for all three predictors. On Open3DSG and SGFN, paired intervals from a cluster bootstrap over scans are strictly positive for Recall and negative for Violation. On VL-SAT, whose Source Recall is already high, the Recall interval contains zero while the Violation interval remains strictly negative, supporting lower Violation without a detectable Recall loss. Open3DSG exposes the failure most strongly: the Source baseline ranks geometrically inconsistent relation candidates highly at small K, whereas RelCompat3D improves Recall and reduces Violation without applying a hard filter. Across $K = 10$–50, no RelCompat3D Recall point estimate is lower than its Source baseline, and the Violation point estimate decreases for every predictor. SGFN changes only marginally at $K = 10$, while Open3DSG exhibits the largest gains throughout this range.

No single comparator dominates across all predictors and reported K values. Under the same ranking procedure, the matched MLP raises Open3DSG Recall more than the RelCompat3D product at $K = 50$ (46.70% versus 44.18%) but also has higher Violation (4.13% versus 3.42%); the two are nearly tied on VL-SAT and SGFN. The comparisons expose predictor-dependent trade-offs: on VL-SAT and SGFN, the RelCompat3D product avoids the larger Recall losses at small K observed with rank fusion. On Open3DSG, the matched MLP raises Recall at the cost of higher Violation. Product (all families) reaches higher aggregate Recall and can lower aggregate Violation, but changes support/contact selections and can mask a support/contact regression. The pooled compatibility ablation is reported in the supplement. The matched MLP, RankAvg, and RRF use the same family-aware re-ranking procedure as RelCompat3D; they therefore preserve the relation-family sequence and keep support/contact candidates in source order. Product (all families) does not impose this restriction.

Surface-Based Geometry Audit

The primary compatibility heads and verifier share some OBB-derived measurements. We therefore remeasure proximity and vertical violations directly from reconstructed instance surfaces. The point estimate uses robust vertex distance and height; the mesh estimate uses area-weighted triangle samples. Thresholds are fixed from training-split positives, and neither estimate reads OBB measurements, pre-

Predictor	Ranking rule	$K=5$		$K=10$		$K=20$		$K=50$		$K=100$	
		R	V	R	V	R	V	R	V	R	V
Open3DSG	Source	3.42	52.05	9.87	32.89	19.89	20.99	40.43	13.87	51.11	12.42
	RelCompat3D	3.73	0.94	11.35	2.33	23.62	3.13	44.18	3.42	56.92	3.24
	Matched MLP	3.70	4.95	11.78	4.97	24.67	4.56	46.70	4.13	59.89	3.71
	RankAvg	3.78	1.09	11.08	2.48	22.41	3.53	43.23	4.58	54.08	5.48
	RRF	3.60	18.99	10.70	16.06	21.63	13.24	42.37	9.48	54.76	7.72
	Product (all families)	9.99	4.05	19.66	4.52	32.85	4.28	46.55	2.65	60.05	3.30
VL-SAT	Source	41.94	0.29	63.22	0.82	80.74	1.42	92.72	2.68	96.35	4.76
	RelCompat3D	42.07	0.15	63.39	0.57	80.82	1.14	92.77	1.97	96.58	2.95
	Matched MLP	42.09	0.15	63.47	0.51	80.92	1.09	92.72	1.89	96.50	2.96
	RankAvg	35.42	0.15	51.76	0.47	69.49	0.99	90.26	1.62	97.00	2.25
	RRF	36.78	0.15	55.49	0.51	74.72	1.04	91.74	1.80	97.10	2.47
	Product (all families)	41.77	0.11	63.32	0.49	80.97	1.10	92.93	2.03	96.88	3.25
SGFN	Source	31.17	2.37	39.75	3.49	49.12	3.22	74.02	3.85	92.35	6.30
	RelCompat3D	31.17	2.37	39.75	3.47	49.14	2.97	74.50	2.63	93.03	3.50
	Matched MLP	31.17	2.37	39.75	3.47	49.19	2.96	74.57	2.58	92.88	3.50
	RankAvg	31.09	2.37	38.80	3.47	46.58	2.95	70.22	2.36	92.80	2.59
	RRF	31.14	2.37	38.92	3.47	47.03	2.97	68.13	2.53	87.74	3.02
	Product (all families)	31.14	2.48	40.61	3.21	51.26	2.74	77.06	2.56	94.18	3.72

Table 1: Exact-match Recall (R↑) and Violation (V↓) computed by a rule-based geometry verifier. All entries are percentages. RelCompat3D, matched MLP, RankAvg, and RRF use the same family-aware ranking procedure; Product (all families) applies compatibility to every relation family.

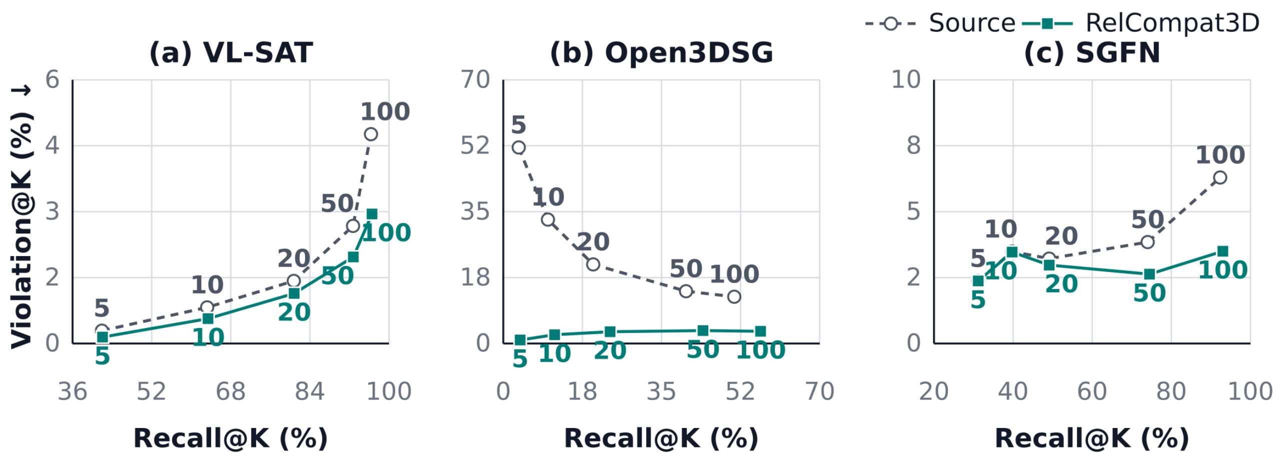

Figure 3: Recall@K–Violation@K trajectories. Each line connects $K \in \{5, 10, 20, 50, 100\}$ for one predictor and ranking rule. Rightward and downward movement is preferred. Numbers identify K for both curves; axis ranges differ by predictor.

dictor identity, the predictor score, learned compatibility, or primary verifier labels.

Table 2 reports strict point–mesh consensus for surface-based Violation@K, treating disagreements as uncertain. This metric is a separate construct and its absolute values are not directly comparable to primary Violation. At $K = 50$ and $K = 100$, RelCompat3D lowers surface-based Violation@K for all predictors, with paired intervals from a cluster bootstrap over scans below zero. The two estimates agree in direction across the full curve except for the tied SGFN $K = 5$ result. Rankings are unchanged, so Recall remains that of Table 1; all budgets, coverage, thresholds, and geometric interventions are in the supplement.

Ablations and Controls

Table 3 shows that wrong-pair and shuffled geometry lower Recall and raise Violation, indicating that ordered-pair identity matters. Wrong-predicate and fixed-predicate endpoint swap are distinct controls, but both reverse the signed vertical relation, partly explaining their nearly identical aggregate Violation. Distance only raises Open3DSG Recall at the cost of substantially higher Violation and underperforms on VL-SAT and SGFN. Compatibility only lowers Violation at the cost of Recall, showing that it and the predictor score are complementary. Removing the exact verifier scalar preserves most gains, whereas removing all related measurements attenuates them; feature removal, linked positive–

Predictor	Source	Ours	Change (95% CI)	Coverage
VL-SAT	16.43	14.33	-2.10 [-2.42, -1.81]	95.5/71.0
Open3DSG	45.96	4.83	-41.13 [-42.84, -39.37]	98.2/83.8
SGFN	11.56	8.13	-3.43 [-3.92, -3.00]	95.8/79.9

Table 2: Surface-based Violation at $K = 50$ (%, ↓) using point–mesh consensus. Change is RelCompat3D minus Source in percentage points with a paired 95% interval from resampled scans; coverage is measured/decidable. Absolute values are not directly comparable to Table 1.

Predictor	Condition	R@50	V@50	R@100	V@100
Open3DSG	**RelCompat3D**	44.18	3.42	56.92	3.24
	Wrong predicate	42.65	22.00	53.47	20.98
	Wrong pair	38.52	8.48	49.32	8.35
	Shuffled geometry	38.44	12.93	48.69	12.43
	Fixed-predicate swap	42.67	22.00	53.73	20.98
	Distance only	51.16	8.24	63.22	9.55
	Compatibility only	42.07	3.42	56.77	3.24
VL-SAT	**RelCompat3D**	92.77	1.97	96.58	2.95
	Wrong predicate	90.43	4.98	94.81	7.96
	Wrong pair	90.53	2.61	94.96	4.73
	Shuffled geometry	89.17	3.06	94.11	5.60
	Fixed-predicate swap	90.43	4.98	94.81	7.96
	Distance only	81.90	5.34	89.80	8.09
	Compatibility only	67.65	1.61	84.09	2.03
SGFN	**RelCompat3D**	74.50	2.63	93.03	3.50
	Wrong predicate	71.55	9.43	89.98	12.99
	Wrong pair	71.58	3.99	87.99	6.65
	Shuffled geometry	70.62	4.74	86.78	8.36
	Fixed-predicate swap	71.55	9.43	90.03	12.99
	Distance only	63.19	10.00	84.06	12.66
	Compatibility only	52.79	2.32	70.72	2.30

Table 3: Ablations and counterfactual controls. All entries are percentages; R is Recall (↑) and V is Violation (↓). Every condition uses the same candidates and family-aware ranking procedure as Table 1.

counterfactual ordering, and transformation checks are in the supplement.

Discussion and Limitations

All three predictors share one 3DSSG target; ReplicaSSG/FROSS shows non-uniform transfer under ontology and geometry shifts (supplement). RelCompat3D assumes known instances and measurable ordered-pair geometry. Support/contact errors, functional or reference-frame-dependent relations, and complete graph generation remain outside scope.

The compatibility targets and primary verifier share some OBB-derived measurements. The surface audit excludes these inputs and labels but shares the reconstructed surface and ontology, so it is not independent physical-validity evidence. Broader claims require independent labels, contact evidence, and datasets.

Conclusion

RelCompat3D excludes predictor scores from compatibility. Across three predictors on one 3DSSG target, it yields higher or tied Recall point estimates and lower Violation point estimates at $K = 10$–50 while keeping support/contact candidates in source order.

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